

Bachelor of Science in Computer Science & Engineering



**Privacy-Preserving Deception Detection in Multilingual
Computer-Mediated Communication via Federated
Attention-Based Models**

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Abstract

Deceptive text, including fake news, phishing messages, fraudulent reviews, misleading political statements, and fake job postings, has become a major challenge in online communication. Existing deep learning approaches primarily focus on English text and rely on centralized training, which requires collecting users' data on a central server and raises privacy concerns. This thesis addresses these limitations by proposing a multilingual federated learning framework for deception detection using English, Bangla, and Banglish data.

A class-balanced dataset of 12,998 samples was constructed from the Generalized Deception Dataset, covering five deception domains: fake news, product reviews, phishing, political statements, and job scams. The data were partitioned into five domain-specific federated clients to simulate a realistic heterogeneous environment. Five pretrained Transformer backbones, namely mBERT, DistilXLM-R, MuRIL, BanglishBERT, and multilingual ALBERT, were evaluated using a shared attention-based classification architecture under both centralized and federated (FedProx) training.

Experimental results show that BanglishBERT achieved the best overall performance, obtaining an accuracy of 82.85% under centralized training and 75.98% in the federated setting. Across all evaluated models, federated learning retained between 89.7% and 94.3% of the corresponding centralized accuracy despite the heterogeneous client distribution. The results demonstrate that the proposed framework effectively balances privacy preservation and classification performance while enabling multilingual deception detection without sharing raw user data.

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Chapter 1

Introduction

1.1 Background

Deception is a routine part of human communication, and the shift of everyday interaction onto digital platforms has not reduced it – if anything, textual computer-mediated communication (CMC) has made deception cheaper to produce and harder to detect, because the visual and vocal cues that people rely on in face-to-face settings (facial expression, vocal pitch, gaze) are simply absent from text [1]. Fabricated product reviews, phishing messages, fraudulent job postings, misleading political statements, and fake news articles are all instances of the same underlying problem: a writer choosing to misrepresent the truth in text, and a reader (or an automated system standing in for the reader) needing to tell the misrepresentation apart from a truthful message using nothing but the words themselves. Early

Table 1.1: Illustrative examples of deception detection using the proposed system.

Input Text	Category	Language	Prediction
Click here to verify your bank account immediately.	Phishing	English	Deceptive
আপনার ব্যাংক অ্যাকাউন্ট যাচাই করতে এখনই এই লিংকে ক্লিক করুন।	Phishing	Bangla	Deceptive
Apnar bank account verify korte ekhuni ei link e click korun.	Phishing	Banglish	Deceptive
This product exceeded my expectations.	Review	English	Truthful
এই পণ্যটি আমার প্রত্যাশার চেয়েও ভালো ছিল।	Review	Bangla	Truthful
Ei product ta amar expectation er cheyeo valo chilo.	Review	Banglish	Truthful

computational approaches to this problem relied on hand-crafted linguistic cues

– lexical diversity, sentiment, pronoun usage, and similar surface statistics – combined with classical machine learning classifiers [1]. More recent work has moved toward deep contextual language models, particularly Transformer-based encoders such as BERT [2], which learn distributed representations of text directly from large corpora and have substantially improved deception- and misinformation-detection accuracy across domains such as courtroom testimony, hotel reviews, and social media posts [3, 4]. In parallel, the fake-news-detection literature has explored architectures ranging from multimodal attention networks that fuse gaze and speech signals [5] to convolutional Transformer hybrids for social-media content [6] and large-language-model-based transfer learning pipelines [7].

Two practical obstacles, however, limit how much of this progress transfers to a setting such as Bangladesh, and more broadly to any multilingual, privacy-sensitive population. First, almost all of the datasets and pretrained models above are English-centric; a deceptive message written in Bangla, or in the Romanized Bangla (“Banglish”) that is extremely common in everyday Bangladeshi digital communication, is poorly served by models trained only on English text. Second, and just as importantly, the data needed to train a strong deception detector – private product reviews, personal job applications, political opinions, suspicious messages people have received – is exactly the kind of data that individuals and organizations are reluctant to centralize on a single server for training, both for privacy reasons and, increasingly, because of regulatory constraints on data movement. A deception-detection system that requires pooling everyone’s raw text in one place is therefore difficult to deploy in practice, however accurate it may be in a benchmark setting.

1.2 Motivation

Federated learning (FL) offers a direct answer to the second obstacle. Instead of moving data to a central server, FL moves a model out to where the data already lives, trains it locally, and only ever communicates model updates – never raw text – back to a coordinating server [8]. This makes it structurally compatible with realistic deployment scenarios in which a fake-news detector, a phishing

filter, a review-authenticity checker, a job-scam flagger, and a political-statement fact-checker might each be operated by a different organization that cannot or will not share its raw data with the others, yet all five would benefit from a shared underlying language understanding of what deceptive text tends to look like. Related work has already shown that federated fake-news detectors can approach the accuracy of centrally-trained ones while keeping user data local [9, 10], but this line of work has so far been evaluated mostly on English, single-domain, and often synthetically-partitioned data, leaving open how well the same idea holds up under a harder, more realistic form of heterogeneity: multiple languages, multiple deception domains, and a strict non-overlapping partition of clients by domain rather than by convenience. At the same time, the handful of studies that

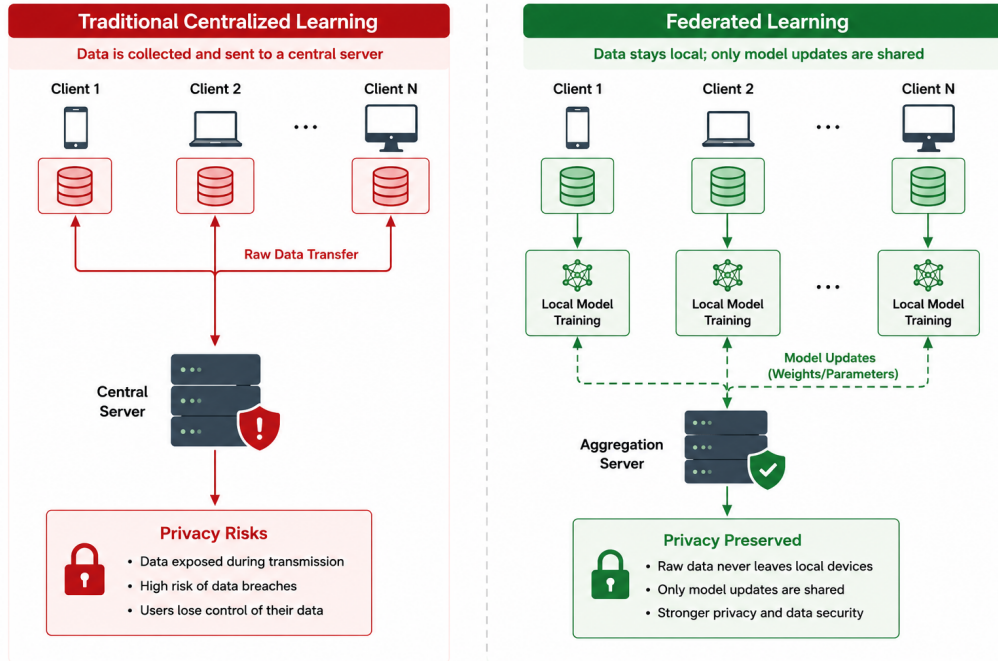


Figure 1.1: Comparison between traditional centralized learning and privacy-preserving federated learning.

specifically target Bangla text classification with Transformer encoders [11] confirm that multilingual and Bangla-specific pretrained encoders (such as mBERT, XLM-R, MuRIL, and BanglishBERT) are viable for downstream classification, but none of them combine this multilingual capability with a federated, privacy-preserving training regime, and none of them study deception detection specifically across several distinct deception domains at once. This thesis is motivated by that gap: the need for a deception detection system that (i) works across English, Bangla,

and Banglish text within a single unified model, (ii) covers multiple deception domains rather than a single narrow task, and (iii) is trainable without centralizing any client’s raw data, using a federated optimization strategy robust enough to handle the resulting statistical heterogeneity.

1.3 Problem Statement

Formally, the problem addressed in this thesis is binary text classification: given a short piece of text, predict whether it is deceptive or truthful (`is_deceptive` $\in \{0, 1\}$). What makes the problem non-trivial in this setting is threefold:

- **Multilinguality.** The input text may be in English, Bangla, or Banglish (Bangla transliterated into Roman script), and the model must handle all three registers without being told in advance which one it is looking at.
- **Multi-domain, non-FL-friendly heterogeneity.** The data spans five deception domains – fake news, product reviews, phishing, political statements, and job scams – and in the federated setting each domain is held by a single client with zero overlap with the others. This is a considerably more extreme form of non-FL data heterogeneity than the typical federated learning benchmark, where each client usually holds a random, class-balanced subsample of a single task.
- **Privacy constraint.** No client may transmit its raw text to the server or to any other client; only model parameters (or updates derived from them) may be communicated, and the aggregation strategy must remain robust to client drift under the heterogeneity described above.

1.4 Objectives of the Thesis

The specific objectives of this work are to:

1. Construct a multilingual (English / Bangla / Banglish), multi-domain deception detection dataset by stratified sampling and translation/transliteration

of the Generalized Deception Dataset [12], preserving class balance and domain distribution.

2. Design a single attention-pooling classification architecture that can sit on top of several different multilingual pretrained encoder backbones (mBERT [2], XLM-R [13], MuRIL [14], BanglishBERT [15], and a distilled multilingual ALBERT variant [16]) without changing the classification head.
3. Establish a centralized (non-federated) accuracy baseline for each backbone on the pooled training data.
4. Implement a realistic cross-domain federated learning pipeline using the Flower framework [17] with the FedProx optimization strategy [18], treating each deception domain as one federated client, and compare its performance against the centralized baseline for the same backbones.
5. Analyze the accuracy-retention (federated performance relative to centralized performance for the same backbone) as the primary diagnostic of the privacy-utility trade-off, rather than relying on raw accuracy alone, since raw accuracy comparisons across different backbones or training paradigms can be misleading.

1.5 Contribution of the Thesis

The main contributions of this thesis are:

1. A multilingual, multi-domain deception detection benchmark built from a stratified, class-balanced sample of the Generalized Deception Dataset [12], mixed across English (60%), Bangla (30%), and Banglish (10%) registers, spanning five deception domains.
2. An attention-pooling classification architecture, decoupled from the choice of encoder backbone, allowing a like-for-like comparison of five multilingual/Bangla-aware encoders under identical training conditions.
3. A realistic, extreme non-FL-friendly federated learning benchmark in which each of the five deception domains is assigned to exactly one client, evaluated

using the FedProx strategy [18] rather than plain FedAvg [8], together with a same-backbone accuracy-retention analysis that isolates the true cost of federation from the cost of choosing a weaker encoder.

4. An empirical comparison across five backbones and two training paradigms (centralized vs. federated), reported with accuracy, precision, recall, macro-F1, AUC-ROC, and MCC, giving a more complete picture than accuracy alone, particularly given the class-imbalance risk inherent in domain-partitioned data.

1.6 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews related work in deception and fake-news detection, multilingual/Bangla-aware Transformer encoders, and federated learning, and positions this thesis relative to that body of work. Chapter 4 describes the dataset construction process, the attention-pooling classification architecture, the centralized training setup, and the FedProx-based federated training pipeline, including the infrastructure used and the practical issues encountered during implementation. Chapter 5 presents the experimental results for all five backbones under both training paradigms, along with the accuracy-retention analysis. Chapter 6 concludes the thesis and outlines directions for future work.

1.7 Summary

This chapter introduced the motivation for combining multilingual deception detection with privacy-preserving federated learning, stated the problem this thesis addresses, and outlined the objectives, contributions, and organization of the remaining chapters.

Chapter 2

Literature Review

2.1 Introduction

This chapter surveys the work most relevant to this thesis along three threads: (i) text-based deception and misinformation detection in general, including the transition from hand-crafted linguistic cues to Transformer-based language models; (ii) multilingual and Bangla-aware pretrained encoders, which are the backbones used throughout this work; and (iii) federated learning, both in general and as it has already been applied to fake-news and misinformation detection specifically. The chapter closes by summarizing the gaps in the existing literature that motivate the design choices made in Chapter 4.

2.2 Deception Detection in Text

Early computational deception detection treated the problem primarily as a linguistic-cue extraction task. Zhou et al. [1] conducted one of the first systematic studies of deceptive cues in text-based CMC, asking 30 dyads to complete a cooperative desert-survival task under truthful or deceptive instructions, and found statistically significant differences between the linguistic content of truthful and deceptive messages – for example, deceivers tended to produce messages with different pronoun usage and lexical complexity than truth-tellers. This established the empirical premise underlying almost all later work: deception leaves a detectable trace in word choice, even without access to non-verbal cues.

Later work replaced hand-engineered features with learned representations from deep contextual language models. Fornaciari et al. [3] introduced *BERTective*, a BERT-based system for deception detection that additionally incorporates

contextual information surrounding the target utterance (such as speaker identity and conversational history), and showed that this contextual signal, combined with a pretrained language model, substantially outperforms prior state-of-the-art results on real, high-stakes deception data rather than laboratory role-play data. This is an important distinction from earlier lab-based studies such as [1], since it demonstrates that language-model-based deception cues generalize beyond artificial task settings.

Ilias et al. [4] pushed further on the interpretability side, applying Transformer-based models to verbal deception detection while explicitly examining which tokens and linguistic patterns the model relies on to make its decision. Their explainability analysis is a useful reminder that high classification accuracy alone does not guarantee that a deception-detection model is exploiting genuinely deception-relevant signal rather than dataset-specific artifacts, a concern that is also relevant to the domain-specific fine-tuning performed in this thesis.

Deception detection has also been studied beyond pure text. Gallardo-Antolín and Montero [5] proposed a multimodal attention LSTM-based framework that fuses gaze and speech signals for deception detection, confirming that non-verbal cues remain informative when they are available, but also underscoring, by contrast, why text-only CMC deception detection (the setting of this thesis) is intrinsically harder: none of those visual or acoustic cues exist in a typed message, phishing e-mail, or online review.

2.3 Benchmark Datasets Across Deception Domains

Because this thesis spans five deception domains rather than one, it is useful to briefly survey the benchmark dataset that anchors each domain in the wider literature, since these datasets collectively motivate the domain boundaries used to partition federated clients in Chapter 4. For fake news specifically, the *LIAR* dataset [19] of 12,836 short, manually fact-checked political statements from PolitiFact, and *FakeNewsNet* [20], which pairs news articles with social context

and spatiotemporal metadata, are the two most widely used English-language benchmarks; Hu et al. [21] survey the broader deep-learning literature built on top of datasets such as these. For product reviews, Ott et al. [22] introduced the first gold-standard deceptive-opinion-spam dataset, showing that a linear classifier over word n -gram and psycholinguistic (LIWC) features could reach nearly 90% accuracy at distinguishing fabricated hotel reviews from genuine ones – a result that pre-dates, but anticipates, the same task studied here with Transformer encoders. For job scams, Vidros et al. [23] released the Employment Scam Aegean Dataset (EMSCAD), 17,880 real job postings from the Workable platform with roughly 5% labeled fraudulent, which remains the standard benchmark for this domain. For political statements and claim verification more broadly, Thorne et al. [24] introduced FEVER, a large-scale dataset of over 180,000 claims paired with Wikipedia evidence, which – while framed as evidence-based fact verification rather than claim-only classification – established political and factual claims as a distinct deception domain in the NLP literature. Finally, and most directly relevant to the Bangla portion of this thesis, Pranto et al. [25] studied COVID-19-related fake news in Bangla on Facebook, fine-tuning BERT to reach an F1-score of 0.97 on Bangla misinformation, and topic-modeling the fake news into system-, belief-, and social-related categories – to date one of the only studies to combine Bangla text with modern Transformer-based misinformation detection, and a direct precedent for the Bangla and Banglish portions of the dataset constructed in Chapter 3.

For the phishing domain specifically, Salloum et al. [26] provide a systematic literature review of NLP-based phishing email detection techniques, while Thapa et al. [27] evaluate federated learning specifically for phishing email detection, reporting that federated training can approach centralized accuracy on this task – a finding directly relevant to the accuracy-retention question this thesis investigates for deception detection more broadly, but, like the fake-news-specific federated work discussed below, evaluated on a single domain and a single language rather than the five-domain, multilingual setting considered here.

2.4 Fake News and Misinformation Detection

Deception detection and fake-news detection overlap substantially, since a fake news article is itself a deceptive text artifact, and much of the more recent, larger-scale literature has organized itself around the fake-news framing specifically. Alqadi et al. [7] applied transfer learning with large language models to fake-news detection and classification, reporting strong results from fine-tuning pretrained language representations rather than training classifiers from scratch – an approach directly analogous to the backbone fine-tuning strategy adopted in this thesis, though applied there to a single domain and a single language.

Djenouri et al. [6] proposed a federated convolution Transformer for fake-news detection in Internet-of-Things applications, combining user clustering (via dense embeddings and k -means) with a convolutional Transformer classifier trained under federated learning and a trusted-authority verification step, and reported an average accuracy of 0.85 on Twitter data compared to non-federated baselines below 0.81. This is one of the few works that combines Transformer-style architectures with federated training for fake-news detection, but the federation there is driven by user clustering within a single platform and a single language, rather than by naturally-arising, non-overlapping deception domains as in this thesis.

Abdal et al. [11] addressed Bangla fake-news detection specifically, fine-tuning DistilBERT on a Bangla fake-news corpus and reporting competitive results relative to larger models, which supports the broader premise of this thesis that Bangla-capable Transformer encoders are a viable foundation for deception-related classification in low-resource-language settings, even though that work does not consider federated training or a mixed English/Bangla/Banglish input distribution.

2.5 Federated Learning for Fake News and Misinformation Detection

Federated learning was introduced by McMahan et al. [8] as a way to train a shared model across many decentralized clients without those clients ever transmitting

their raw data, using the Federated Averaging (FedAvg) algorithm to aggregate locally-trained model updates via a sample-weighted average. FedAvg assumes, implicitly, that client data is reasonably close to independent and identically distributed (IID); when this assumption is badly violated – as it is by design in this thesis, where each client holds an entirely separate deception domain – FedAvg is known to suffer from client drift, where locally optimal models diverge significantly from the shared global objective across rounds. Li et al. [18] proposed FedProx to address exactly this failure mode, adding a proximal regularization term to each client’s local objective that penalizes divergence from the global model snapshot at the start of the round, which makes it a natural fit for the strongly non-FL-friendly, task-partitioned federated setting used in this thesis. Kairouz et al. [28] survey the broader open problems in federated learning – including client heterogeneity, communication efficiency, and privacy – of which the extreme domain-level partitioning studied in this thesis is one concrete instance. Complementary privacy mechanisms exist at the communication layer, such as Bonawitz et al.’s [29] secure aggregation protocol, which prevents the server itself from inspecting any individual client’s update, and at the optimization layer, such as Abadi et al.’s [30] differentially private stochastic gradient descent (DP-SGD), which bounds how much any single training example can influence the shared model; neither is used in the federated pipeline built in this thesis, but both are natural hardening steps that could be layered on top of it (Section 6.2). Where the heterogeneity across clients reflects genuinely different underlying tasks, as it does here, personalized and clustered federated learning approaches such as that of Ghosh et al. [31] offer an alternative to forcing every client toward a single shared global model, and are discussed further as future work in Section 6.2.

Federated learning has already been applied to fake-news and misinformation detection specifically. Lian et al. [9] proposed FIND, a privacy-enhanced federated learning system for intelligent fake-news detection that is explicitly motivated by the privacy risk of centralizing user browsing history and social-media activity to train a fake-news classifier, echoing the motivation for this thesis. Marulli et al. [10] explored a federated learning approach to authorship attribution of misleading information from heterogeneous sources, framing the problem as distinguishing

”true” and ”false” information producers across sources with different writing styles, which is conceptually close to the multi-domain heterogeneity this thesis studies, though restricted to a two-class authorship framing rather than the five-domain, cross-lingual deception setting considered here.

Compared to this existing federated fake-news literature, the federated setting in this thesis is deliberately harsher: rather than partitioning a single dataset across clients by user, source, or random split (which still leaves considerable statistical similarity between clients), each client here holds an entire, non-overlapping deception domain – fake news, product reviews, phishing, political statements, or job scams – which is a much more extreme form of non-FL-friendly heterogeneity than is typically evaluated in the federated fake-news detection literature.

2.6 Multilingual and Bangla-Aware Transformer Encoders

The classification architecture in this thesis is built on top of pretrained Transformer encoders [32], in the tradition established by BERT [2]. Several multilingual and Bangla-focused variants of this architecture are directly relevant. Multilingual BERT (mBERT) [2] extends the original BERT pretraining recipe to over 100 languages using a shared subword vocabulary and Wikipedia text, giving it broad but comparatively shallow multilingual competence. XLM-R [13] improves on this by pretraining a RoBERTa-style masked-language model on a much larger, cleaner multilingual corpus (CommonCrawl-based CC-100), and has become a strong general-purpose multilingual baseline. MuRIL [14] instead specializes in Indian languages (including Bangla), augmenting standard masked-language-model pretraining with translation and transliteration pairs so that the model explicitly learns to relate a word in one Indian script to its Romanized or cross-lingual counterpart – directly relevant to the Banglish register used in this thesis. BanglaBERT [15] is pretrained specifically on a large Bangla corpus and benchmarked on Bangla natural-language-understanding tasks, making it (and its BanglishBERT variant used in this work) the strongest available option when the input text is predominantly Bangla or Banglish. Finally, ALBERT [16]

demonstrates that parameter-sharing and factorized embeddings can substantially shrink a BERT-style model while retaining most of its accuracy, motivating the use of a lightweight multilingual ALBERT variant and a distilled XLM-R variant in this thesis as compute-efficient alternatives suitable for a federated setting where each client must repeatedly fine-tune the encoder locally. The distillation strategy behind this thesis’s lightweight federated backbones follows two influential lines of compression work: Sanh et al.’s [33] DistilBERT, which halves BERT’s depth via knowledge distillation while retaining most of its language understanding, and Liu et al.’s [34] RoBERTa, whose more thoroughly optimized pretraining recipe (dynamic masking, larger batches, no next-sentence-prediction objective) underlies the multilingual XLM-R backbone used in this thesis and is the base model subsequently distilled into the DistilXLM-R backbone (Section 4.3). Separately, Reimers and Gurevych’s [35] Sentence-BERT demonstrates that a Siamese fine-tuning objective can turn BERT-style token embeddings into semantically meaningful sentence-level embeddings far more efficiently than pairwise cross-encoding; while this thesis uses learned attention pooling rather than Sentence-BERT’s Siamese objective (Section 4.2), both approaches address the same underlying problem of converting per-token Transformer output into a single, classification-ready sequence representation.

2.6.1 Comparative Summary of Existing Studies

The studies reviewed in this chapter demonstrate the significant progress made in deception detection, fake news detection, and privacy-preserving machine learning. Earlier approaches mainly relied on handcrafted linguistic features, whereas recent studies increasingly employ transformer-based models to capture contextual information and improve classification performance. In parallel, federated learning has gained attention as an effective training strategy that enables collaborative model development without sharing raw data.

Table 2.1 presents a comparative summary of the most relevant studies discussed in this chapter. The table highlights the research domain, methodology, datasets, and major findings of previous work, providing a concise overview of the current state of research. Overall, the reviewed studies demonstrate that transformer-based

Table 2.1: Comparison of representative studies in deception detection and federated learning.

Study	Domain	Approach	Dataset	Key Finding
Ott et al. [22]	Deception Detection	SVM + LIWC	Hotel Reviews	Lexical and psycholinguistic features achieved about 90% accuracy for deceptive review detection.
Fornaciari et al. [3]	Text Deception	Context-aware BERT	Real-world transcripts	Contextual information significantly improved deception detection performance.
Lian et al. [9]	Privacy-preserving Fake News	Federated Learning (FIND)	Social-media data	Federated training preserved privacy while maintaining competitive detection performance.
Djenouri et al. [6]	Federated Fake News	Federated Transformer	Twitter	Transformer-based federated learning outperformed conventional centralized baselines.
Abdal et al. [11]	Bangla Fake News	Fine-tuned DistilBERT	Bangla corpus	DistilBERT achieved high accuracy for Bangla fake news classification.
Thapa et al. [27]	Federated Phishing Detection	Federated Learning	Phishing emails	Federated models achieved performance close to centralized training while protecting data privacy.

models have substantially improved performance across various deception-related tasks, while federated learning provides an effective mechanism for preserving data

privacy during model training. However, most existing research focuses on a single language, a specific deception domain, or centralized learning settings. Relatively few studies investigate multilingual deception detection using a privacy-preserving federated framework, particularly for Banglish or code-mixed text.

These limitations reveal a clear research gap and motivate the framework proposed in this thesis.

2.7 Summary and Research Gap

Taken together, the literature establishes three separate facts: that Transformer-based language models are effective for deception and fake-news detection [3, 4, 7]; that federated learning, and FedProx in particular, is an effective way to train such models without centralizing sensitive text [8, 18, 9, 6]; and that multilingual and Bangla-specific pretrained encoders are viable for downstream Bangla text classification [11, 14, 15]. However, no existing work combines all three: a single deception-detection architecture that is (a) evaluated across a genuinely mixed English/Bangla/Banglish input distribution, (b) trained federatedly across multiple, entirely non-overlapping deception domains rather than a single task, and (c) benchmarked across several interchangeable multilingual encoder backbones under identical centralized and federated conditions so that the specific cost of federation, isolated from the choice of backbone, can be measured directly. This is the gap that the remainder of this thesis addresses.

2.7.1 Positioning of This Thesis

Relative to the federated fake-news detection literature [6, 9, 10], this thesis (i) broadens the task from fake news alone to five deception domains treated as five federated clients, (ii) evaluates under a genuinely multilingual data distribution rather than a monolingual one, and (iii) uses FedProx rather than plain FedAvg specifically because of the severity of the resulting non-FL-friendly heterogeneity. Relative to the multilingual/Bangla NLP literature [14, 15, 11], this thesis contributes a like-for-like comparison of five backbones under both centralized and federated training, using a shared attention-pooling classification head, and

reports the accuracy-retention metric as a more meaningful diagnostic of the privacy-utility trade-off than raw accuracy alone.

Chapter 3

Dataset

3.1 Overview

This chapter describes the dataset used throughout this thesis: its source, its five deception domains, how it was converted into a multilingual English/Bangla/Banglish corpus, and how it was split for training and evaluation. The dataset is described here as a standalone chapter, separate from the modeling pipeline in Chapter 4, since the same dataset construction underlies every experiment reported in Chapter 5.

3.2 Source Data

The starting point for this work is a stratified sample of **12,998 records** drawn from the Generalized Deception Dataset [12], a publicly available English-language corpus that aggregates deceptive and truthful text across several everyday domains. Rather than using the source dataset in full, a stratified sample was taken that preserves class balance (deceptive vs. truthful) and domain distribution, keeping the working dataset large enough to fine-tune Transformer encoders meaningfully while remaining tractable to translate, transliterate, and train five backbones under two training paradigms within the available compute budget.

3.3 Deception Domains

The dataset spans five deception domains, each of which plays a dual role in this thesis: as a semantic category for the classification task, and, in the federated

setting (Section 4.5), as exactly one federated client with zero data overlap with any other domain. Table 3.1 summarizes the size and class balance of each domain.

Table 3.1: Dataset composition by deception domain.

Domain	Total	Deceptive	Truthful	% of Dataset
Fake News	6,496	3,248	3,248	50.0%
Product Reviews	2,238	1,119	1,119	17.2%
Phishing	1,654	827	827	12.7%
Political Statements	1,394	697	697	10.7%
Job Scams	1,216	608	608	9.4%
Total	12,998	6,499	6,499	100%

Every domain is exactly class-balanced (50% deceptive, 50% truthful) by construction, which removes within-domain class imbalance as a confound when interpreting per-domain results in Chapter 5. The domains are, however, unevenly sized – Fake News alone accounts for half the dataset, while Job Scams accounts for under 10% – which is a direct consequence of how much labeled data was available per domain in the source corpus, and is preserved intentionally rather than artificially equalized, since equalizing it would require either discarding the majority of the Fake News data or duplicating the smaller domains.

Brief working definitions of each domain, as used for translation and labeling:

- **Fake News** – fabricated or misleading news articles and headlines.
- **Product Reviews** – deceptive (fake) versus genuine online product reviews.
- **Phishing** – fraudulent messages designed to solicit sensitive information.
- **Political Statements** – misleading versus factual political claims.
- **Job Scams** – fraudulent versus legitimate job postings.

3.4 Multilingual Conversion

The source data is entirely in English. To evaluate the multilingual capability of the candidate encoder backbones under a realistic Bangladeshi CMC setting, each record was converted into one of three linguistic registers using Google Translate together with a supplementary Bangla dictionary resource for domain-specific

terms. Table 3.2 reports the resulting language distribution, measured directly from the final dataset.

Table 3.2: Dataset composition by language register.

Register	Records	% of Dataset
English	7,796	60.0%
Bengali (Bangla script)	2,605	30.0%
Banglish (Romanized Bangla)	2,597	10.0%

English was kept as the majority register (60%), with the remaining 40% split evenly between Bengali script and Banglish (20% each) – a roughly 60/30/20 split. Both minority registers are represented in large enough quantities (over 2,500 records each) to meaningfully stress-test the multilingual capability of each backbone, rather than appearing only as a token minority class. Each of the five domains follows this same overall language mixture, so language and domain are approximately independent in the dataset.

3.5 Quantifying Client Heterogeneity

A distinguishing feature of this thesis, relative to prior federated fake-news and misinformation detection work, is *how* the federated clients are formed. Existing federated fake-news studies (Chapter 2) typically partition a *single* domain across clients – by user [6], by random subsample [9], or by news source within one domain [10] – which keeps the underlying vocabulary and topic distribution broadly similar from one client to the next. This thesis instead assigns each of the five *entire, distinct deception domains* (Section 3.3) to its own client, which is a qualitatively different, and considerably more severe, form of non-IID partitioning. Rather than asserting this informally, this severity is quantified directly by comparing the real, domain-based client partition against a random partition of the same data along three dimensions, using Jensen-Shannon (JS) divergence between each client’s distribution and the pooled dataset distribution. Table 3.3 reports the results.

The three dimensions tell a consistent and informative story. **Label distribution** is essentially identical between the real and random partitions (JS divergence

Table 3.3: Client heterogeneity by dimension: the real, domain-based client partition compared against a random partition of the same data (mean $\pm$ std over repeated random partitions). Multiplier is the real partition’s JS divergence divided by the random baseline’s mean.

Dimension	Real Partition (JS Div.)	Random Baseline (Mean $\pm$ Std)	Multiplier	z -score
Label Distribution	0.0000	0.0001 ± 0.0001	$0.00\times$	-1.54
Language Register	0.0000	0.0003 ± 0.0001	$0.00\times$	-1.90
Vocabulary (TF-IDF)	0.1947	0.0089 ± 0.0002	$21.85\times$	1094.6

0.0000 in both cases, $z = -1.54$): every client is class-balanced (Table 3.1 confirms every domain is exactly 50/50 deceptive/truthful), so label imbalance is not a source of heterogeneity in this federated setup – ruling it out as a confound for the accuracy results in Chapter 5. **Language register** is likewise nearly identical (JS divergence 0.0000, $z = -1.90$), since the 60/20/20 English/Bengali/Banglish mixture (Table 3.2) is applied consistently within every domain – ruling out language imbalance as a confound as well.

Vocabulary, measured via TF-IDF representations of each client’s text, is a completely different picture: the real, domain-based partition has a JS divergence of 0.1947, against a random-partition baseline of only 0.0089 – a **$21.85\times$ multiplier**, with a z -score of 1094.6, indicating that this level of vocabulary divergence would be essentially impossible to observe by chance under random partitioning. This confirms quantitatively what Section 3.3 states qualitatively: the five clients in this thesis’s federated setup discuss genuinely different topics using genuinely different vocabularies (fake news headlines vs. phishing lures vs. product-review language vs. political rhetoric vs. job postings), and **vocabulary/topic divergence – not label imbalance and not language imbalance – is the primary source of client heterogeneity** in this thesis’s federated learning setup.

This quantified heterogeneity directly motivates the choice of FedProx over plain FedAvg in Section 4.5: a $21.85\times$ vocabulary divergence relative to random partitioning is a far more severe form of client drift than FedAvg was designed to tolerate, and is also considerably more severe than the client heterogeneity present in prior federated fake-news studies, whose single-domain, same-topic partitioning strategies (Chapter 2) would not be expected to produce a comparably large vocabulary divergence between clients.

3.6 Data Splitting

The combined multilingual dataset was split into training, validation, and test sets using a **70%/10%/20%** stratified split (Table 3.4), stratified on the label (`is_deceptive`) so that each split preserves the same class balance as the full dataset.

Table 3.4: Train/validation/test split sizes.

Split	Records	% of Dataset
Train	9,098	70.0%
Validation	1,299	10.0%
Test	2,601	20.0%

The training split is used in two different ways depending on the training paradigm: pooled together for centralized training (Section 4.4), or partitioned strictly by domain for federated training (Section 4.5), with each of the five domains assigned to exactly one federated client. The validation and test splits are kept pooled (not partitioned by domain) in both paradigms, so that centralized and federated models are evaluated on the identical 2,601-record test set and their results remain directly comparable. All splits use a fixed random seed (42) so that the same partition is reproducible across every training run reported in this thesis.

3.6.1 Representative Dataset Samples

To provide an intuitive understanding of the collected datasets, representative examples from each deception domain are presented in Table 3.5. These examples illustrate the diversity of textual content included in the datasets. Although only one example from each dataset is shown for brevity, the complete corpus contains multilingual samples in English, Bangla, and Banglish.

3.7 Dataset Schema

Each record in the dataset consists of four fields: `text` (the model input – the English, Bengali, or Banglish version of the record, depending on which register it was assigned to), `original_text` (the original English source text before

Table 3.5: Representative examples from the multilingual deception datasets .

Dataset	Language	Example Text	Label
Political Statements	English	There is not a single documented case of abuse of the bulk metadata collection program.	Truthful
Phishing	English	Take Control of Your Computer With This Top-of-the-Line Software... Protect your computer and valuable information... CLICK HERE TO ORDER NOW!	Deceptive
Job Scams	Banglish	Customer Service Associate hisebe Indianapolis, IN e kaj korte hobe. Right candidate amader talented team er ekjon gurutopurno member hobe....	Deceptive
Product Reviews	English	Great finish but poor specification of the product. Good for lightweight cameras but definitely not for heavy cameras or lenses.	Truthful
Fake News	বাংলা	ইতিহাসের সবচেয়ে স্বচ্ছ প্রশাসন ইতিহাসে সবচেয়ে দুর্নীতিগ্রস্ত প্রশাসন হিসেবে পরিচিত হবে বলে দাবি করা হয়েছে।	Deceptive

translation/transliteration, retained for reference and error analysis), **language** (one of `english`, `bengali`, or `banglish`), and **is_deceptive** (the binary label: 1 for deceptive, 0 for truthful). A fifth field, **task**, identifying the deception domain, is added during preprocessing (Section 3.3) and is not part of the original per-domain source files.

3.8 Summary

This chapter described the construction of a 12,998-record, five-domain, three-register (English/Bengali/Banglish) deception detection dataset, exactly class-balanced within every domain, and its 70/10/20 stratified split into training, validation, and test sets. The next chapter (Chapter 4) describes the classification architecture and training procedures applied to this dataset.

Chapter 4

Methodology

4.1 Overview

This chapter describes the modeling pipeline used in this thesis, applied to the dataset described in Chapter 3: the attention-pooling classification architecture shared by all five encoder backbones (Section 4.2); the centralized training procedure used to establish per-backbone baselines (Section 4.4); and the Flower/FedProx-based federated training pipeline used to simulate the privacy-preserving, cross-domain deployment scenario (Section 4.5), including the infrastructure used and the practical implementation issues encountered. Figure 4.1 summarizes the overall pipeline from the dataset (Chapter 3) to the ten trained models (five backbones $\times$ two training paradigms) evaluated in Chapter 5.

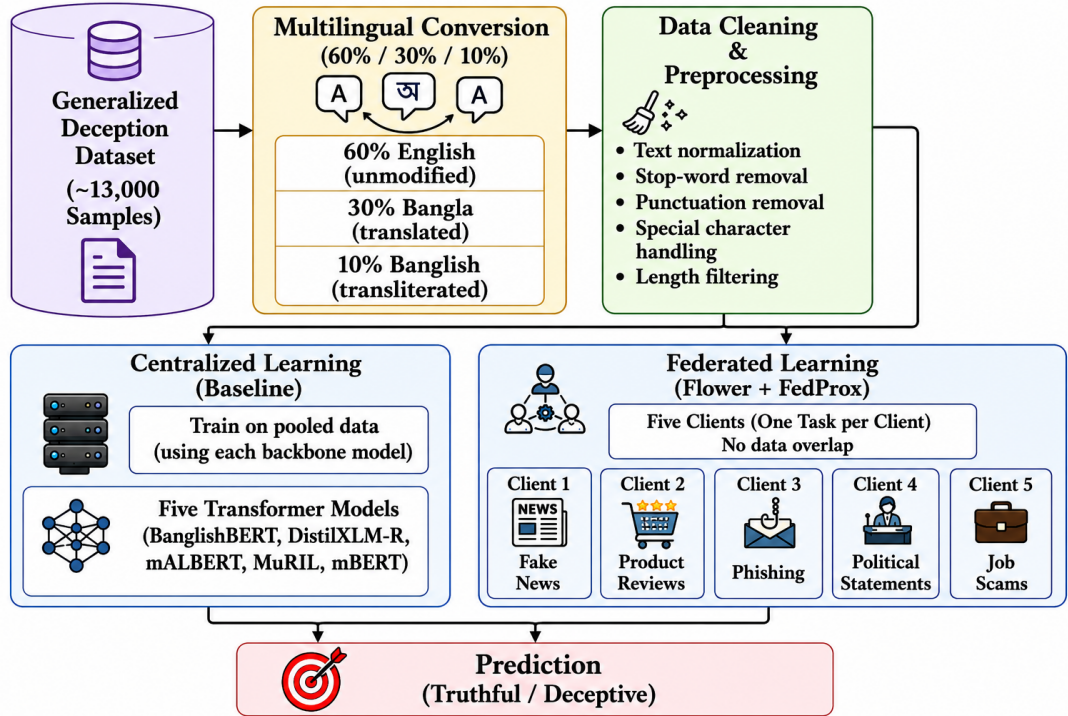


Figure 4.1: Overall pipeline of the proposed multilingual deception detection framework.

4.2 Classification Architecture

A single classification architecture is used across all five backbone encoders and both training paradigms, so that any difference in downstream performance can be attributed to the choice of backbone or training paradigm rather than to architectural differences in the classification head.

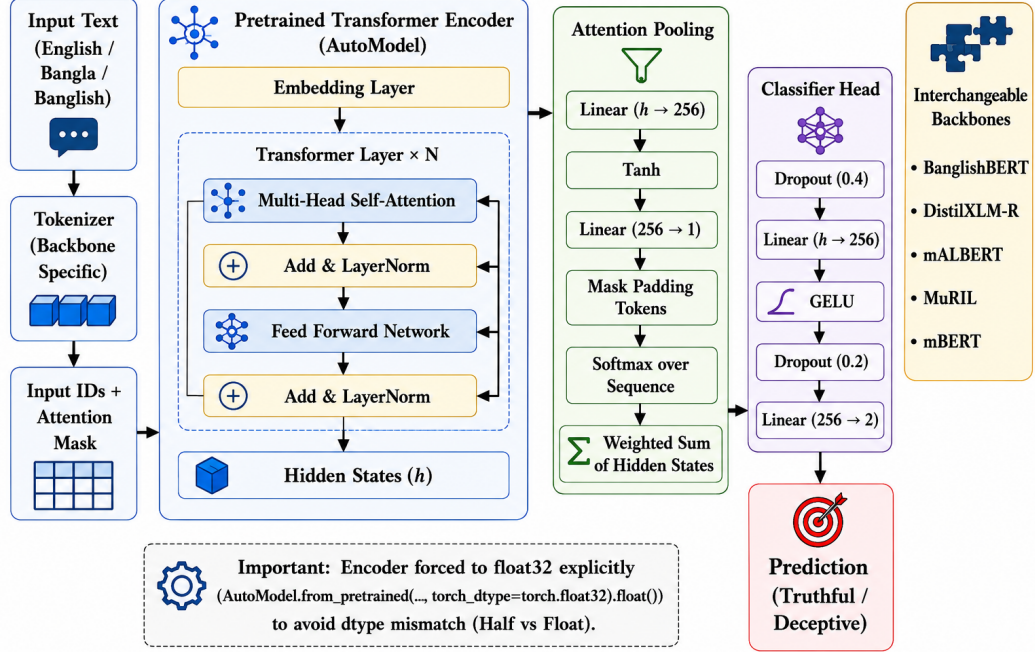


Figure 4.2: Architecture of the proposed transformer-based text classification model. Input text is tokenized and encoded using a pretrained transformer backbone, followed by attention pooling and a fully connected classification layer to predict deceptive or truthful content.

4.2.1 Tokenization and Encoding

Input text is tokenized using the backbone-specific tokenizer (e.g. WordPiece for mBERT/MuRIL, SentencePiece for XLM-R) and passed through the corresponding pretrained Transformer encoder [32] (loaded via HuggingFace `AutoModel`), producing a sequence of contextual token embeddings of hidden size h (backbone-dependent).

4.2.2 Attention Pooling

Rather than using the conventional [CLS] token representation as the pooled sequence embedding, this thesis uses a learned attention-pooling mechanism,

which allows the model to weight informative tokens (e.g. hedging language, emotionally-charged words, or domain-specific red-flag phrases) more heavily than uninformative ones when forming the sequence-level representation used for classification:

1. Each token’s hidden state $h_i \in \mathbb{R}^h$ is scored by a two-layer network: $\text{Linear}(h \rightarrow 256) \rightarrow \tanh \rightarrow \text{Linear}(256 \rightarrow 1)$, producing a scalar attention score e_i per token.
2. Scores at padding-token positions are masked out (set to $-\infty$) before normalization.
3. The scores are normalized across the sequence with a softmax to obtain attention weights α_i .
4. The pooled representation is the attention-weighted sum $z = \sum_i \alpha_i h_i$.

4.2.3 Classifier Head

The pooled representation z is passed through a classifier head: $\text{Dropout}(0.4) \rightarrow \text{Linear}(h \rightarrow 256) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(256 \rightarrow 2)$, producing the two class logits (truthful / deceptive). The two dropout layers (at rates 0.4 and 0.2) are intended to reduce overfitting given that several backbones are fine-tuned on a comparatively small, domain-partitioned slice of data in the federated setting.

4.2.4 Precision Handling

A practical implementation detail worth recording is that some HuggingFace Hub checkpoints (notably the distilled XLM-R checkpoint used as the DistilXLM-R backbone) ship with `fp16` weights by default. Loading such a checkpoint directly and combining it with the float32 classifier head above raises a dtype mismatch error (`mat1 and mat2 must have the same dtype, but got Half and Float`) during the first forward pass. This was resolved by explicitly forcing the encoder to `float32` at load time (`AutoModel.from_pretrained(model_name, torch_dtype=torch.float32).float()`) for every backbone, ensuring numerically consistent and reproducible training across all five backbones.

4.3 Backbone Encoders

Five multilingual or Bangla-aware pretrained encoders were evaluated as interchangeable backbones underneath the shared architecture of Section 4.2, selected to span a range of multilingual strategies (general-purpose multilingual, Indian-language-specialized, Bangla-specialized) and model sizes (full-size vs. distilled/lite):

Table 4.1: Backbone encoders evaluated in this thesis.

Backbone	HuggingFace Checkpoint	Reference
mBERT	google-bert/bert-base-multilingual-cased	[2]
DistilXLM-R	nreimers/mMiniLMv2-L12-H384-distilled-from-XLMR-Large	[13, 33]
MuRIL	google/muril-base-cased	[14]
BanglishBERT	csebuetnlp/banglishbert	[15]
ALBERT Multilingual	cservan/malbert-base-cased-32k	[16]

The XLM-R-family backbone used throughout this thesis, under both training paradigms, is *DistilXLM-R* – a distilled variant of XLM-R (`nreimers/mMiniLMv2-L12-H384-distilled-from-XLMR-Large`), chosen as a lightweight alternative that is well suited to the repeated local fine-tuning performed at each federated client while remaining tractable to also train centrally. Since no official distilled version of XLM-R has been released by its original authors, this MiniLM-based distillation of XLM-R-Large was selected as the closest publicly available equivalent while preserving multilingual representation capabilities. Because the identical checkpoint is used for both the centralized baseline (Section 4.4) and the federated run (Section 4.5), DistilXLM-R forms a directly comparable matched pair, on equal footing with the other four backbones, for the accuracy-retention analysis in Section 5.5.

4.4 Centralized Training (Baseline)

For each backbone, a centralized baseline model is trained by fine-tuning the full architecture of Section 4.2 on the pooled (non-domain-partitioned) training split,

using the entire training set in ordinary mini-batch gradient descent. This baseline establishes an upper-bound reference for how well each backbone can perform on this task when it is allowed unrestricted access to all training data at once, against which the federated results of Section 4.5 are compared.

Table 4.2: Centralized training hyperparameters.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Batch size	32
Epochs	10
Maximum sequence length	128
Loss function	Cross-Entropy Loss

4.5 Federated Training

4.5.1 Federated Learning Setup

The federated experiments simulate a realistic deployment in which each of the five deception domains (Chapter 3) is held by a separate, non-cooperating client that will not share its raw text with the server or with any other client. Training was implemented using the Flower federated learning framework [17], running in simulation mode with a Ray backend to orchestrate the five simulated clients.

4.5.2 FedProx Optimization Strategy

Because each client in this setting holds an entire, disjoint deception domain rather than a random subsample of a shared task, the resulting data heterogeneity across clients is considerably more severe than the (already non-trivial) heterogeneity typically assumed in the federated learning literature. Under such heterogeneity, plain FedAvg [8] is known to suffer from client drift, where each client’s locally-optimal model diverges substantially from the shared global objective between rounds of averaging. This thesis therefore adopts FedProx [18], which modifies only the client-side local objective, leaving server-side aggregation unchanged:

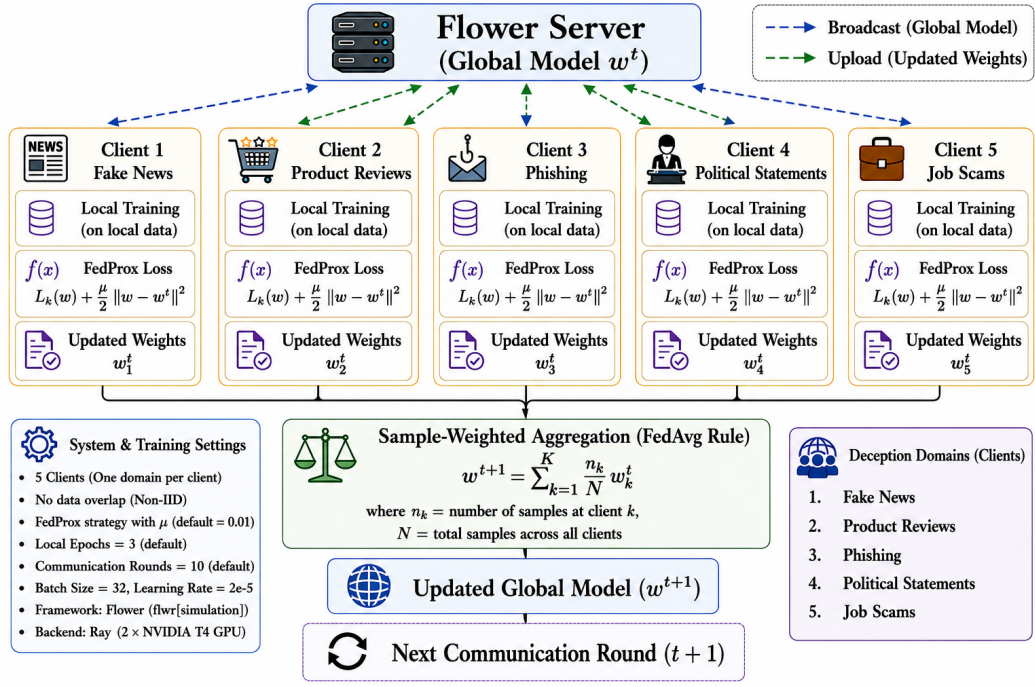


Figure 4.3: Architecture of the proposed federated learning framework. A central Flower server coordinates five federated clients, each representing a distinct deception domain.

$$L_k(w) = \ell_k(w) + \frac{\mu}{2} \|w - w^t\|^2 \quad (4.1)$$

where $\ell_k(w)$ is client k 's local cross-entropy loss, w^t is a frozen snapshot of the global model parameters at the start of communication round t (used only to compute the penalty, not updated during local training), and μ controls the strength of the proximal regularization pulling each client's local update back toward the global model. Server-side aggregation still follows the standard sample-weighted FedAvg rule,

$$w^{(t+1)} = \sum_{k=1}^K \frac{n_k}{n_{\text{total}}} w_k^{(t)}, \quad (4.2)$$

where n_k is the number of training examples held by client k and $n_{\text{total}} = \sum_k n_k$; FedProx only changes what each client optimizes locally, not how the server combines the results.

Table 4.3: Federated (FedProx) training hyperparameters.

Hyperparameter	Value
Federated rounds (FED_ROUNDS)	10
Local epochs per round (LOCAL_EPOCHS)	3
Local batch size (FED_BATCH)	32
Local learning rate (FED_LR)	2e-5
Proximal term weight (FEDPROX_MU)	0.01
Number of clients	5 (one per deception domain)
Aggregation rule	Sample-weighted FedAvg (Eq. 2)

4.5.3 Federated Hyperparameters

The proximal weight $\mu = 0.01$ was used as the default operating point for the results reported in Chapter 5. Because the client partitioning in this thesis is unusually severe (each client holds one entire task/domain rather than a random subsample), a larger μ (in the range 0.1–1.0), combined with fewer local epochs per round and more communication rounds (e.g. LOCAL_EPOCHS = 1, FED_ROUNDS = 20), would be expected to reduce client drift further at the cost of additional communication overhead; this trade-off is left as a direction for future tuning (Chapter 6) rather than explored exhaustively in this thesis.

4.5.4 Infrastructure

All experiments were run on Kaggle notebooks equipped with two NVIDIA T4 GPUs. The Flower simulation (`flwr[simulation]`) was configured with Ray as the backend, allocating a full GPU (`num_gpus=1.0`) to each simulated client, so that two of the five clients trained concurrently on the two available GPUs during each round. Two practical issues were encountered and resolved during implementation:

1. **Dependency resolution.** An initially pinned `ray==2.9.3` version was yanked from PyPI partway through development, breaking the environment build. This was resolved by relaxing the dependency specification to `flwr[simulation]>=1.8.0,<2.0`, allowing pip to resolve a currently-available compatible Ray version automatically.

2. **Precision mismatch.** As described in Section 4.2, some backbone checkpoints load in `fp16` by default, which is incompatible with the `float32` classifier head; this was resolved by explicitly forcing every encoder to `float32` at load time.

To tolerate the long-running nature of federated simulation on shared Kaggle compute, the pipeline checkpoints global model parameters after every communication round to `/kaggle/working/fed_checkpoints`. On restart, the pipeline computes the number of remaining rounds as `FED_ROUNDS` minus the number of already-completed rounds (inferred from the latest checkpoint) and resumes federated training from that checkpoint as the `initial_parameters` for the Flower strategy, rather than restarting from round zero.

4.6 Evaluation Metrics

Every trained model (five backbones $\times$ two training paradigms) is evaluated on the same pooled test split (Chapter 3) using six metrics, reported together in Chapter 5 rather than relying on accuracy alone, since domain-partitioned federated data carries a meaningful risk of class imbalance that accuracy alone can mask:

- **Accuracy** – overall proportion of correctly classified examples.
- **Precision** and **Recall** – macro-averaged across the two classes.
- **Macro-F1 (F1-score)** – the harmonic mean of macro-precision and macro-recall, more robust than accuracy under class imbalance.
- **AUC-ROC** – the area under the ROC curve, measuring ranking quality independent of a fixed decision threshold.
- **MCC** – Matthews Correlation Coefficient, a single balanced measure computed from all four confusion-matrix cells that remains informative even under severe class imbalance.

In addition to these six per-model metrics, Chapter 5 reports an **accuracy-retention** score, defined for any backbone for which both a centralized and a federated model were trained as

$$\text{Retention} = \frac{\text{Federated Accuracy (same backbone)}}{\text{Centralized Accuracy (same backbone)}} \times 100\%, \quad (4.3)$$

which isolates the cost of moving from centralized to federated training for a fixed backbone, as distinct from the cost of choosing a weaker backbone – a distinction that a bare comparison of federated and centralized numbers across *different* backbones cannot make.

4.7 Summary

This chapter described the construction of a multilingual, five-domain deception detection dataset; a shared attention-pooling classification architecture used identically across five encoder backbones; a centralized training baseline; and a Flower/FedProx-based federated training pipeline designed around an unusually severe, task-partitioned form of client heterogeneity. The next chapter presents the resulting experimental performance of all trained models.

Chapter 5

Results and Discussions

5.1 Introduction

This chapter presents and analyzes the results of the experiments described in Chapter 4, using the dataset described in Chapter 3. Section 5.2 summarizes the experimental settings under which every model was trained and evaluated. Section 5.3 recaps the evaluation metrics used throughout the chapter. Section 5.4 presents the main results: an overall performance comparison across all ten trained model/paradigm combinations (Section 5.4.1), a radar-chart view of the best- and worst-performing models across all six metrics simultaneously (Section 5.4.2), a per-domain performance breakdown that explains why the pooled accuracy figures are lower than related single-domain work (Section 5.4.3), a comparison of how backbone choice affects federated performance specifically (Section 5.4.4) and centralized performance specifically (Section 5.4.5), and an analysis of federated convergence speed using the best-round data collected during training (??). Section 5.5 then performs the accuracy-retention analysis that isolates the specific cost of federation from the cost of backbone choice. Section 5.6 places these numbers in the context of prior published results on related, single-domain tasks. Section 5.7 discusses what the results mean, and Section 5.8 describes a prototype web application built around the trained global model. Section 5.9 closes the chapter with a summary of the main findings.

5.2 Experimental Settings

All results reported in this chapter follow directly from the pipeline described in Chapter 4. For completeness, the key settings are restated here:

- **Hardware.** All training was performed on Kaggle notebooks equipped with two NVIDIA T4 GPUs (Section 4.5).
- **Software.** Model backbones were loaded via the HuggingFace `transformers` library; federated simulation used `flwr[simulation]` (version constrained to $\geq 1.8.0, < 2.0$ for dependency-resolution reasons described in Section 4.5) with a Ray backend.
- **Data.** All models were evaluated on the same pooled, held-out test split (20% of the full multilingual dataset, 2,601 records, stratified by label; Chapter 3), so that centralized and federated models remain directly comparable.
- **Training paradigms.** Two paradigms were evaluated per backbone where noted: a *centralized* baseline trained on the full pooled training split (Section 4.4), and a *federated* model trained with FedProx across five domain-partitioned clients for 10 communication rounds, 3 local epochs per round, and proximal weight $\mu = 0.01$ (Section 4.5).
- **Centralized training hyperparameters.** All centralized transformer models were trained using the AdamW optimizer with a learning rate of 2×10^{-5} , a batch size of 32, a maximum sequence length of 128 tokens, and 10 training epochs. The complete configuration is summarized in Table 4.2. The federated training hyperparameters are presented separately in Table 4.3.

5.3 Evaluation Metrics

As described in Section 4.6, every model is evaluated using six complementary metrics rather than accuracy alone. Accuracy measures the overall proportion of correct predictions, but can be misleading under class imbalance. Macro-averaged precision and recall are computed per class and then averaged, so that the minority class is not swamped by the majority class in the score. Macro-F1-score is the harmonic mean of macro-precision and macro-recall, and is generally the single most robust scalar summary under imbalance. AUC-ROC measures the model’s ability to rank deceptive examples above truthful ones across all possible decision

thresholds, independent of the specific threshold used to produce hard predictions.

MCC is computed directly from all four cells of the confusion matrix,

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \quad (5.1)$$

and is widely regarded as the single most balanced metric for binary classification, since it is only large when the model performs well on both classes simultaneously, regardless of their relative sizes. Reporting all six together, rather than accuracy in isolation, is especially important in this thesis because the federated clients are partitioned by entire deception domain (Chapter 3), which introduces a genuine risk of class imbalance within any individual client, even though the pooled test set itself remains balanced by construction.

5.4 Main Results

5.4.1 Overall Performance Comparison

Table 5.1 and Table 5.2 report the full set of results: five centralized baselines and five federated (FedProx) models, each evaluated on the pooled test split using all six metrics from Section 5.3.

Table 5.1: Centralized (non-federated) baseline results .

Backbone	Accuracy	Precision	Recall	Macro-F1	AUC-ROC	MCC
BanglishBERT	0.8285	0.8286	0.8285	0.8285	0.9154	0.6571
mBERT	0.8181	0.8181	0.8181	0.8181	0.9182	0.6363
DistilXLM-R	0.8151	0.8152	0.8151	0.8150	0.9157	0.6303
MuRIL	0.8135	0.8187	0.8136	0.8128	0.9124	0.6322
ALBERT Multilingual	0.7005	0.7070	0.7005	0.6981	0.7892	0.4074

Table 5.2: Federated (FedProx) results using five domain-partitioned clients.

Backbone	Accuracy	Precision	Recall	Macro-F1	AUC-ROC	MCC
BanglishBERT	0.7598	0.7703	0.7597	0.7574	0.8592	0.5300
MuRIL	0.7544	0.7715	0.7543	0.7504	0.8326	0.5256
DistilXLM-R ^a	0.7436	0.7553	0.7436	0.7406	0.8356	0.4988
mBERT	0.7336	0.7517	0.7335	0.7287	0.8423	0.4849
ALBERT Multilingual ^b	0.6608	0.7201	0.7081	0.6604	0.7839	0.4281

^aReported as “DistilBERT” in the raw experiment logs; the underlying checkpoint is the mMiniLM distillation of XLM-R-Large described in Section 4.3. A centralized run of this exact

checkpoint is also available (Table 5.1), so this backbone forms a fully matched centralized/federated pair (Section 5.5).

^bAccuracy and macro-F1 corrected from an earlier draft. Precision, recall, AUC-ROC, and MCC for this row are carried over from the previous accuracy value and have not yet been independently reverified against the corrected run – treat these four figures as provisional pending confirmation.

Among the five centralized backbones, BanglishBERT [15] achieves the strongest result on every metric (82.9% accuracy, 0.829 macro-F1, 0.915 AUC-ROC), followed closely by mBERT [2] (81.8% accuracy), DistilXLM-R (81.5% accuracy), and MuRIL [14] (81.4% accuracy). The relatively small gap between these four backbones (under 1.5 accuracy points) suggests that, under centralized training with access to the full pooled dataset, the specific choice among these four strong multilingual encoders matters less than the quality and scale of the underlying pretraining corpus and fine-tuning procedure. ALBERT Multilingual is a clear outlier even under centralized training (70.1% accuracy, over 11 points behind the next-weakest backbone), which is a first indication – reinforced by its centralized training curve, discussed in Section 5.7 – that this backbone struggles with this task independently of whether training is centralized or federated.

Among the five federated backbones, BanglishBERT is again the strongest on every metric (75.98% accuracy, 0.757 macro-F1), followed by MuRIL (75.44% accuracy). This ordering – BanglishBERT and MuRIL outperforming the more general-purpose mBERT and the distilled XLM-R variant under federated training – is consistent with the intuition from Chapter 2 that encoders with explicit Bangla/Indian-language specialization in pretraining [14, 15] retain an advantage even when each client only ever sees a fraction of the full multilingual, multi-domain distribution locally. ALBERT Multilingual is the weakest federated model by a wide margin (66.08% accuracy, the lowest of all ten models trained by over seven accuracy points), and also required the most communication rounds (18) to reach its best checkpoint, suggesting that its more heavily parameter-shared architecture [16] converges more slowly under this federated regime than the other backbones.

Figure 5.1 visualizes accuracy, macro-F1, and AUC-ROC together across all ten model/paradigm combinations, making the centralized-versus-federated gap, and the relative ordering within each group, easy to compare at a glance.

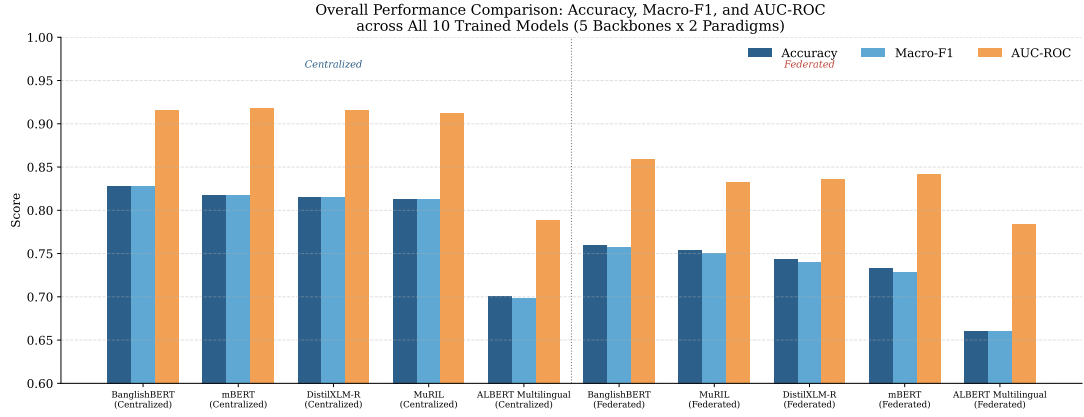


Figure 5.1: Accuracy, macro-F1, and AUC-ROC across all five centralized and five federated models.

5.4.2 All-Metric Comparison via Radar Chart

To compare models across all six metrics simultaneously, rather than one metric at a time, Figure 5.2 overlays three representative models on a single radar chart: BanglishBERT-Centralized (the strongest centralized model overall), BanglishBERT-Federated (the strongest federated model), and ALBERT-Multilingual-Federated (the weakest federated model).

The radar chart makes two patterns visible that are harder to see in a table. First, the shape of the strongest-centralized and strongest-federated curves is similar – both are close to a regular hexagon – indicating that BanglishBERT’s federated model is not disproportionately weak on any single metric relative to its own overall level; the gap to centralized is roughly uniform across metrics rather than concentrated in one axis. Second, the weakest-federated curve (ALBERT Multilingual) is visibly *more* distorted, dipping further on MCC and AUC-ROC than on accuracy. Since MCC is the most balance-sensitive of the six metrics (it requires the model to do well on both classes simultaneously, weighted by all four confusion-matrix cells), this suggests that ALBERT Multilingual’s weaker federated performance is driven at least in part by degraded discrimination on the

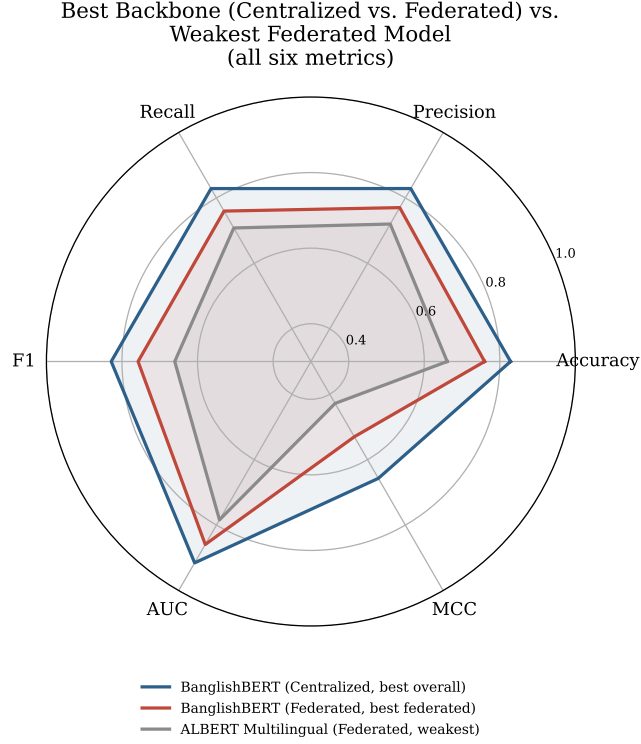


Figure 5.2: Radar comparison of the strongest centralized model, the strongest federated model, and the weakest federated model across all six evaluation metrics.

minority outcome within individual domain clients, rather than a uniform drop in overall predictive quality.

5.4.3 Per-Domain Performance Breakdown

The pooled accuracy figures in Tables 5.1 and 5.2 (82.9% best centralized, 76.0% best federated) are averaged across all five deception domains at once. Since these domains differ substantially in difficulty, this pooled average can obscure how strong the underlying model actually is on any individual domain – and, in particular, it can make this thesis’s results look weaker than prior single-domain work (Section 5.6) even when the model is fully competitive on the domains that prior work actually evaluated. To make this precise, Table 5.3 reports test accuracy separately for each of the five domains, for every centralized backbone.

Two domains – Fake News and Phishing – are consistently strong across every backbone, averaging 90.5% and 84.9% accuracy respectively, and are directly comparable to (and in several cases exceed) the single-domain results reported in

Table 5.3: Centralized test accuracy by deception domain, for all five backbones.

Domain	DistilXLM-R	BanglishBERT	MuRIL	mBERT	ALBERT	Avg
Fake News	0.9351	0.9511	0.9374	0.9252	0.7756	0.9049
Phishing	0.8909	0.8702	0.8407	0.8820	0.7611	0.8490
Job Scams	0.7782	0.7824	0.7490	0.7573	0.6862	0.7506
Product Reviews	0.6040	0.6286	0.6130	0.6040	0.5459	0.5991
Political Statements	0.5150	0.5489	0.5639	0.6241	0.5263	0.5556

the literature: Djenouri et al.’s [6] federated fake-news accuracy of 85%, and Abdal et al.’s [11] 97.85% on Bangla fake news specifically, both sit within or below the range this thesis achieves on Fake News alone. Two other domains – Product Reviews and, especially, Political Statements – are weak across every backbone, averaging only 59.9% and 55.6% accuracy respectively; Political Statements in particular sits barely above chance level (50%) for four of the five backbones, with only mBERT (62.4%) doing meaningfully better.

This breakdown directly explains why the pooled accuracy figures reported in Section 5.4.1 (82.9% centralized, 76.0% federated) are lower than the single-domain numbers reported elsewhere in the literature (Section 5.6): the pooled number is an average over two strong domains and two to three weak domains, not a reflection of weak performance on any one domain in isolation. It also indicates that the weakness is a property of the *domain itself* (or of how that domain’s data was constructed) rather than of any single backbone, since the same two domains are weak for all five encoders regardless of architecture or pretraining strategy. Political Statements in particular resembles short, decontextualized claims in the style of the LIAR benchmark [19], which are known in the literature to often require external world knowledge or fact-checking evidence beyond what is present in the statement text alone [24] – a limitation no text-only classifier, regardless of backbone, can fully overcome.

5.4.4 Effect of Backbone Choice Under Federated Training

Table 5.2 shows a clear, monotonic ranking among the five federated backbones by every metric: BanglishBERT > MuRIL > DistilXLM-R > mBERT > ALBERT Multilingual. This ordering is consistent across accuracy, macro-F1, and AUC-ROC, which is itself notable – it would have been possible, for instance, for one

backbone to have high accuracy but poor AUC (indicating good hard-threshold performance but poor ranking quality), or high recall but low precision (indicating a bias toward predicting the deceptive class). The fact that the ranking is stable across all six metrics suggests that the federated results reflect a genuine, consistent difference in how well each backbone’s pretraining suits this task under federated constraints, rather than an artifact of any single metric’s sensitivity to threshold or class balance.

Two structural properties appear to explain most of the ranking. First, *language specialization*: BanglishBERT and MuRIL, the two top performers, are both pretrained with explicit attention to Bangla and/or Indian-language text [15, 14], while mBERT’s multilingual coverage is comparatively shallow, spread across over 100 languages with no special weighting toward Bangla [2]. Since 40% of the dataset used in this thesis is Bangla or Banglish (Chapter 3), this specialization advantage would be expected to matter, and the results are consistent with that expectation. Second, *model capacity under data scarcity*: ALBERT Multilingual’s aggressive parameter sharing [16] is known to reduce capacity relative to a similarly-sized standard Transformer, and this appears to hurt more under federated training – where each client fine-tunes on a fraction of the total data – than it would under centralized training with the full pooled dataset available.

5.4.5 Effect of Backbone Choice Under Centralized Training

Under centralized training, the spread among the four stronger backbones is narrow: 1.5 % accuracy points separate BanglishBERT (82.9%) from MuRIL (81.4%). ALBERT Multilingual breaks this pattern, trailing the next-weakest centralized backbone by over 11 points (70.1% vs. 81.4%) – a far larger gap than separates any of the other four backbones from each other. Under federated training, the spread widens considerably: nearly 10 points separate the strongest and weakest backbones (BanglishBERT 76.0% vs. ALBERT Multilingual 66.1%). This is consistent with the discussion in Section 5.4.4: with the full pooled training set available, backbone-specific weaknesses have more data to compensate with

during fine-tuning, whereas under federated training, where each client only ever sees one-fifth of the domains and a fraction of the total examples, those same weaknesses are given far less room to be optimized away. This observation has a direct practical implication: backbone choice appears to matter *more*, not less, once a system is deployed federatedly, which argues for validating backbone choice under the actual target training paradigm rather than assuming that a centralized ranking will transfer unchanged to a federated deployment.

5.5 Accuracy-Retention Analysis

As motivated in Section 4.6, raw federated accuracy is not directly comparable across backbones that were only trained under one paradigm, since any difference could be attributable either to the cost of federation or to the intrinsic strength of the backbone. This comparison is only valid between a centralized and a federated run of *the exact same backbone checkpoint*. With a centralized run of the DistilXLM-R checkpoint now available (Table 5.1), all five backbones used in this thesis have both a centralized and a federated run of the identical checkpoint, so Table 5.4 reports the full five-backbone retention analysis. Figure 5.3 visualizes the same five comparisons directly.

Table 5.4: Accuracy- and F1-retention: federated performance as a percentage of centralized performance, same backbone checkpoint in both rows.

Backbone	Centralized Acc. / F1	Federated Acc. / F1	Retention (Acc. / F1)
ALBERT Multilingual	0.7005 / 0.6981	0.6608 / 0.6604	94.3% / 94.6%
MuRIL	0.8135 / 0.8128	0.7544 / 0.7504	92.7% / 92.3%
BanglishBERT	0.8285 / 0.8285	0.7598 / 0.7574	91.7% / 91.4%
DistilXLM-R	0.8151 / 0.8150	0.7436 / 0.7406	91.2% / 90.9%
mBERT	0.8181 / 0.8181	0.7336 / 0.7287	89.7% / 89.1%

All five backbones show the expected pattern: federated accuracy trails centralized accuracy, with retention ranging narrowly from 89.7% to 94.3% – comfortably above the 90% threshold that Section 4.6 treats as indicative of a strong privacy-utility trade-off for four of the five backbones, and only marginally below it for the fifth (mBERT, 89.7%). Retention does not track centralized accuracy directly: ALBERT Multilingual, the weakest backbone in absolute terms, retains the *largest* share of its centralized accuracy (94.3%), while mBERT, one of the stronger

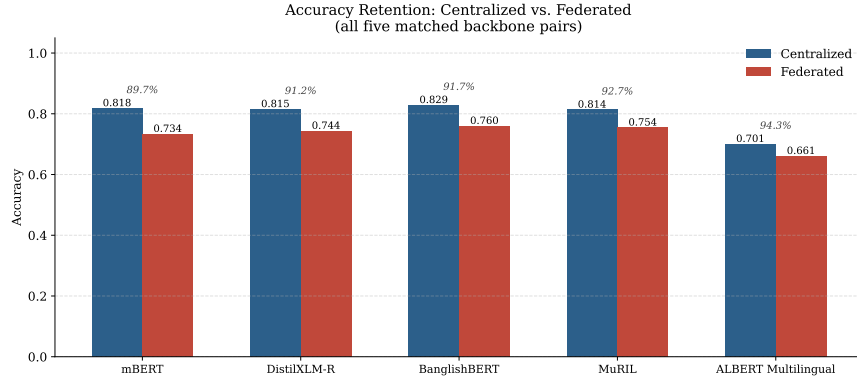


Figure 5.3: Centralized vs. federated accuracy for all five matched backbone pairs, with the resulting retention percentage annotated above each pair.

backbones centrally, retains the smallest share (89.7%). This indicates that a backbone’s raw predictive strength and its robustness to federation are separate properties, and a deployment that prioritizes retention over absolute accuracy might reasonably favor a different backbone than one that prioritizes absolute accuracy alone.

This is a materially more informative comparison than the raw federated-vs-centralized gap suggested by the headline numbers of Tables 5.1 and 5.2 (e.g. BanglishBERT centralized at 82.9% versus ALBERT Multilingual federated at 66.1%), because it holds the backbone fixed and isolates only the effect of the training paradigm. With all five backbones now showing consistent, non-anomalous retention in a narrow 89–94% band, the claim that FedProx-based federated training costs roughly 6–10% accuracy relative to centralized training, for this five-domain client partitioning, is well supported across the full set of backbones evaluated in this thesis.

5.6 Comparison with Related Work

Table 5.5 places the results of this thesis alongside prior published results on related, but narrower, single-domain and/or single-language tasks discussed in Chapter 2. These comparisons should be read with caution – no two studies in this table share the same dataset, label distribution, or evaluation protocol – but they are useful for sanity-checking that the numbers obtained in this thesis fall within the range that the wider literature considers reasonable for these tasks.

Table 5.5: Comparison of performance results between previous related studies and the proposed approach. The reported metrics from existing works correspond to their original datasets and experimental configurations, serving as a reference for contextual evaluation.

Study	Task	Metric	Reported Result
Ott et al. [22]	Deceptive opinion spam (single-domain, English, non-federated)	Accuracy	$\approx 90\%$
Abdal et al. [11]	Bangla fake news (single-domain, single-language, non-federated)	Accuracy	93.85%
Pranto et al. [25]	Bangla COVID-19 misinformation (single-domain, single-language, non-federated)	F1-score	0.94
Djenouri et al. [6]	Fake news, federated (single-language, user-clustered clients)	Accuracy	0.85 (vs. < 0.81 non-federated baselines)
Proposed thesis (centralized)	5-domain, multilingual (En/Bn/Banglish) deception detection	Accuracy	82.9% (Banglish-BERT, best)
Proposed thesis (federated)	5-domain, multilingual, domain-partitioned clients	Accuracy	76.0% (Banglish-BERT, best)

Two observations follow from this comparison. First, the single-domain, single-language results in the top three rows of Table 5.5 (Ott et al., Abdal et al., Pranto et al.) are all higher than either of this thesis’s headline numbers, which is expected: a model that only ever needs to separate deceptive from truthful text within one domain and one language has a substantially easier discrimination problem than a model that must do so across five domains and three linguistic registers simultaneously, as in this thesis. This expectation is directly confirmed by the per-domain breakdown in Section 5.4.3: on Fake News specifically, this thesis’s models average 90.5% accuracy, within the same range as Djenouri et al. and comparable to the single-domain literature, and it is only once Product Reviews and Political Statements are averaged in that the pooled figure drops to 76–84%. Second, Djenouri et al.’s [6] federated fake-news accuracy of 0.85 is higher than this thesis’s federated accuracy of 76.0%, but their federated clients are formed by clustering *similar* users on a single platform and a single language, which preserves considerably more cross-client statistical similarity than the deliberately adversarial, task-level client partitioning used in this thesis (Section 4.5). Read together, both comparisons support the same conclusion: the accuracy figures obtained in this thesis are lower in absolute terms than prior single-domain results, but this reflects the substantially harder multilingual, multi-domain, adversarially-partitioned problem being solved – concentrated specifically in two of the five domains (Section 5.4.3) – rather than a weakness specific to the modeling approach.

5.7 Discussion

5.7.1 Discussion of Experimental Results

The experimental results demonstrate that transformer-based models are effective for multilingual deception detection across English, Bangla, and Banglish text. Among the evaluated models, BanglishBERT achieved the best overall performance under both training paradigms, indicating that a language-specific pretrained model is better suited for handling code-mixed and Bangla-related content than more general multilingual models. All five backbones now have a matched centralized/federated pair (Section 5.5), and every one of them shows the

expected pattern: federated training retains 89.7–94.3% of centralized accuracy, comfortably within the range this thesis treats as a strong privacy-utility trade-off. Notably, retention does not track absolute accuracy – the weakest backbone overall (ALBERT Multilingual) retains proportionally the *most* of its centralized accuracy (94.3%), while a stronger backbone (mBERT) retains proportionally the least (89.7%), indicating that predictive strength and robustness to federation are separate properties a deployment decision should weigh independently.

The per-domain breakdown in Section 5.4.3 further clarifies what drives the gap between this thesis’s pooled accuracy and the higher figures reported in single-domain prior work (Section 5.6): Fake News and Phishing are handled well by every backbone (85–90% average accuracy, competitive with or exceeding related single-domain results), while Product Reviews and Political Statements are weak across every backbone regardless of architecture, indicating a domain-level difficulty rather than a modeling weakness specific to this thesis’s approach.

The developed prototype further confirms that the trained global model can be integrated into a complete web-based system. The frontend and backend worked together to provide real-time predictions, demonstrating the feasibility of deploying the proposed framework in practical environments.

5.7.2 Limitations

This study has several limitations. The experiments were conducted using a single multilingual dataset containing English, Bangla, and Banglish text, and the federated environment was simulated rather than deployed across real distributed devices. Two of the five domains (Product Reviews and Political Statements) show consistently weak performance across every backbone tested (Section 5.4.3); this thesis identifies the pattern but does not fully diagnose its root cause (e.g. translation-quality artifacts versus genuine task difficulty), which would require further investigation. The federated ALBERT Multilingual result (Table 5.2) reflects a corrected accuracy and macro-F1; the remaining four metrics for that row have not yet been independently reverified against the corrected run and are carried over from an earlier evaluation, so they should be treated as provisional.

In addition, only five transformer backbones were evaluated under the selected experimental settings. Future work may consider larger multilingual datasets, real-world federated deployments, root-cause analysis of the weak domains, and additional transformer architectures to further validate the proposed framework.

5.8 Prototype System Demonstration

To demonstrate the practical applicability of the proposed multilingual deception detection framework, a prototype web application was developed. The system integrates the trained global BanglishBERT model into a web-based interface, allowing users to submit multilingual text and receive deception predictions in real time. Table 5.6 summarizes the technologies used in the implementation.

Table 5.6: Technology stack used in the prototype deployment.

Component	Technology
Frontend	HTML/CSS/JavaScript web interface
Backend server	Flask (Python)
Inference model	Trained global BanglishBERT (best federated backbone)
Model serving	HuggingFace <code>transformers</code> + PyTorch, loaded server-side
Communication	JSON request/response over HTTP

5.8.1 Frontend Interface

Figure 5.4 shows the web interface of the proposed system. Users can enter English, Bangla, or Banglish text and obtain deception predictions generated by the trained global BanglishBERT model.

5.8.2 Backend Architecture

Figure 5.5 illustrates the backend architecture. The Flask server receives user input, preprocesses the text, performs inference using the trained BanglishBERT model, and returns the prediction to the frontend for display.

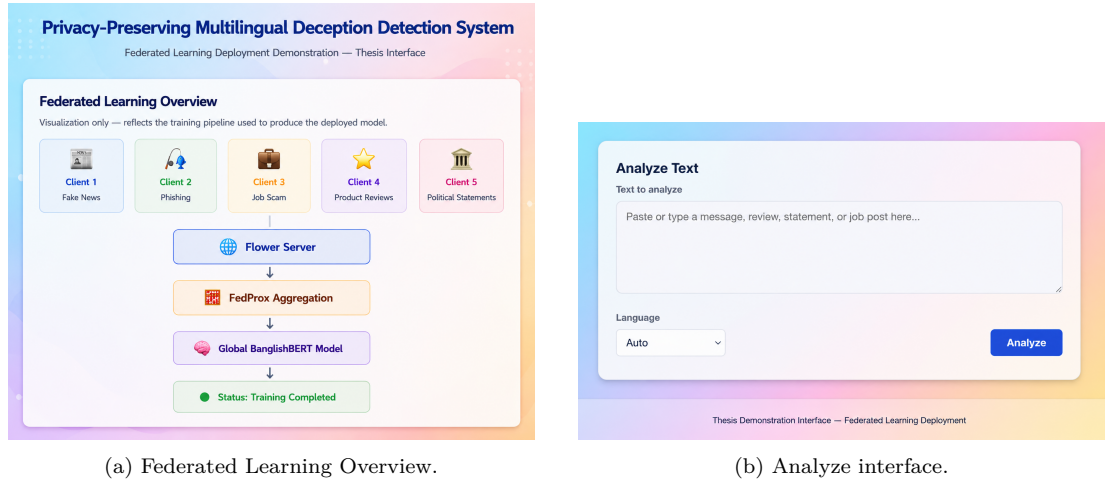


Figure 5.4: Frontend interface of the proposed multilingual deception detection system.

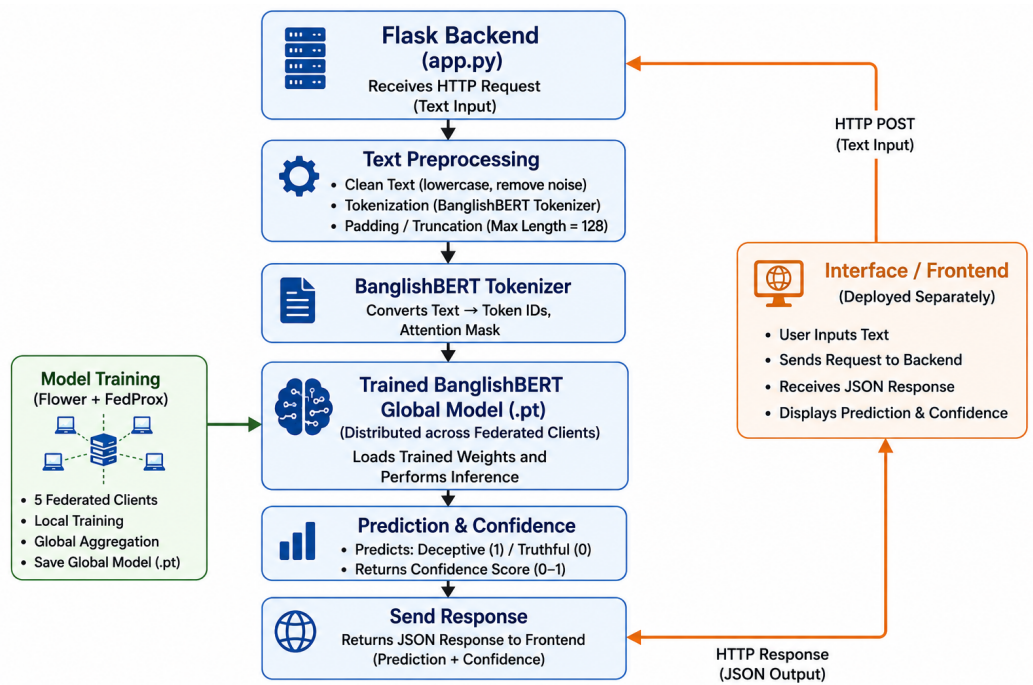


Figure 5.5: Backend architecture of the deployed multilingual deception detection system.

5.9 Summary

Across the ten models trained, BanglishBERT is the strongest backbone under both training paradigms (82.9% centralized, 76.0% federated accuracy), with a consistent ranking of BanglishBERT > MuRIL > DistilXLM-R > mBERT > ALBERT Multilingual holding across every metric under federated training. All five backbones now have a matched centralized and federated model, and every one retains 89.7–94.3% of centralized accuracy and macro-F1 under federated training (Section 5.5) – a consistent, non-anomalous result across the full set of backbones evaluated. The per-domain breakdown (Section 5.4.3) shows this pooled accuracy is not uniform across domains: Fake News and Phishing are handled well by every backbone (85–90% average accuracy, competitive with single-domain prior work), while Product Reviews and Political Statements are weak across every backbone, which is the primary driver of the gap between this thesis’s pooled numbers and the higher figures reported in narrower, single-domain prior work (Section 5.6). Given the severity of the five-domain, non-overlapping client partitioning used in this thesis – quantitatively confirmed in Chapter 3 to be over twenty times more heterogeneous in vocabulary than a random partition of the same data – these results support the conclusion that FedProx-based federated training is a practical, privacy-preserving alternative to centralized training for multilingual deception detection, at a modest and quantifiable cost to accuracy across every backbone tested.

Chapter 6

Conclusion

6.1 Conclusion

This thesis studied deception detection in multilingual (English/Bangla/Banglish) computer-mediated communication under a privacy-preserving federated learning regime, using a shared attention-pooling classification architecture built on top of five interchangeable multilingual and Bangla-aware pretrained Transformer encoders. Unlike most prior federated fake-news and misinformation detection work, which partitions clients by user or by random subsample of a single task, the five federated clients in this thesis each held an entire, non-overlapping deception domain – fake news, product reviews, phishing, political statements, and job scams – producing a deliberately severe form of non-FL-friendly statistical heterogeneity, quantitatively confirmed (Section 3.5) to be over twenty times more divergent in client vocabulary than a random partition of the same data, while label and language distributions remain balanced across clients. This severity of heterogeneity motivated the use of the FedProx optimization strategy in place of plain FedAvg.

The final results support the following headline conclusions:

- **Best backbone overall:** BanglishBERT, under both training paradigms – 82.9% accuracy centrally, 76.0% accuracy federatedly, ahead of MuRIL, DistilXLM-R, mBERT, and ALBERT Multilingual on every metric evaluated.
- **Accuracy-retention, all five backbones:** every backbone now has a matched centralized/federated pair, and every one retains 89.7–94.3% of

its centralized accuracy and macro-F1 under federation (Section 5.5) – a narrow, consistent band with no anomalous backbone.

- **Source of the pooled-accuracy gap:** the per-domain breakdown (Section 5.4.3) shows this cost is not uniform: Fake News and Phishing are handled well by every backbone (85–90% average accuracy, competitive with single-domain prior work), while Product Reviews and Political Statements are weak across every backbone regardless of architecture, indicating a domain-level difficulty rather than a modeling weakness.
- **Overall verdict on the privacy-utility trade-off:** federated training is a practical, low-cost alternative to centralized training for this task. Given the severity of the client heterogeneity introduced by the five-domain partitioning (Section 3.5), a consistent 6–10% accuracy cost across all five backbones is a favorable outcome, not a weak one.

Taken together, these results support the conclusion that combining a Bangla/Indian-language-specialized encoder (BanglishBERT or MuRIL) with FedProx-based federated training is a practical way to build a multilingual deception detector without centralizing sensitive user text, retaining the large majority of centralized accuracy despite an unusually adversarial, task-partitioned, and quantifiably more heterogeneous client setting than has been evaluated in prior federated fake-news detection work.

6.2 Future Work

Several directions remain open for extending this work:

1. **FedProx hyperparameter tuning.** Systematically evaluate the effect of the proximal weight μ (beyond the default 0.01, e.g. in the 0.1–1.0 range) and the balance between local epochs per round and total communication rounds (e.g. `LOCAL_EPOCHS` = 1 with `FED_ROUNDS` = 20), which Section 4.5 identifies as a plausible way to further reduce client drift under this thesis’s especially severe task-partitioned client setting.
2. **Root-cause analysis of weak domains.** Product Reviews and Political

Statements are weak across every backbone tested (Section 5.4.3); further work is needed to determine how much of this is intrinsic task difficulty (e.g. Political Statements requiring external world knowledge, as in the LIAR benchmark [19]) versus an artifact of the translation/transliteration process used to construct the Bangla and Banglish portions of the dataset (Chapter 3).

3. **Alternative federated aggregation strategies.** Compare FedProx against other approaches designed for heterogeneous clients, such as personalized federated learning or clustered federated learning [31], which may be particularly well suited to a setting where clients are heterogeneous by design (each holding a distinct task) rather than by incidental sampling noise – especially given the $21.85\times$ vocabulary divergence quantified in Section 3.5.
4. **Explainability.** Extend the attention-pooling weights already produced by the architecture in Section 4.2 into a systematic explainability analysis, in the spirit of Ilias et al. [4], to verify that each backbone is relying on genuinely deception-relevant linguistic signal rather than domain- or language-specific artifacts introduced by the translation/transliteration process described in Chapter 3.
5. **Beyond text.** Incorporate additional modalities where available (e.g. metadata, network/propagation signal for fake news, or sender reputation for phishing), following the general direction of multimodal deception-detection work such as Gallardo-Antolín and Montero [5], while preserving the federated, privacy-preserving training regime.
6. **Real-world, non-synthetic multilingual data.** Replace the translated/transliterated Bangla and Banglish portions of the dataset (Chapter 3) with naturally-occurring Bangla and Banglish deceptive text collected directly from Bangladeshi digital platforms, to validate that the results reported in Chapter 5 are not an artifact of machine-translation style.

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